

MILDRED LAKE, AB, Canada

WMO: 712550

Lat: 57.0403N

Lon: 111.5592W

Elev: 310

StdP: 97.66

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-34.1	-31.5	-37.7	0.1	-33.9	-35.1	0.1	-31.4	9.3	-7.1	8.2	-7.6	1.4	100	0.648

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	29.2	17.3	27.2	16.4	25.4	15.7	18.8	25.5	17.9	24.4	17.0	23.0	3.8	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.6	12.3	20.7	15.6	11.5	19.8	14.6	10.8	19.2	54.6	25.6	51.6	24.5	48.8	23.0	23.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.5	7.4	6.5	-36.7	32.7	2.8	1.6	-38.7	33.9	-40.3	34.8	-41.9	35.7	-43.9	36.9
			-36.7	20.7	2.7	1.5	-38.7	21.7	-40.3	22.6	-41.8	23.4	-43.8	24.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.0	-16.8	-13.2	-6.3	3.3	10.8	16.1	18.6	16.8	11.3	3.7	-6.8	-14.5	(6)
	DBStd	13.93	9.02	7.87	8.38	6.04	5.05	3.67	3.04	3.56	4.39	4.82	6.57	8.40	(7)
	HDD10.0	3749	830	648	506	209	50	3	0	1	37	202	504	759	(8)
	HDD18.3	6053	1088	882	764	452	237	84	34	72	214	454	754	1017	(9)
	CDD10.0	827	0	0	0	8	74	186	266	211	76	7	0	0	(10)
	CDD18.3	88	0	0	0	0	4	17	42	23	2	0	0	0	(11)
	CDH23.3	921	0	0	0	2	63	191	381	241	41	1	0	0	(12)
	CDH26.7	219	0	0	0	0	14	43	96	58	9	0	0	0	(13)
	WSAvg	3.0	2.6	2.7	3.0	3.5	3.4	3.2	3.1	3.0	3.2	3.2	2.9	2.6	(14)
	PrecAvg	419	20	19	19	19	35	63	74	61	44	27	19	21	(15)
(16) Precipitation	PrecMax	533	35	63	37	65	96	129	137	117	101	152	33	43	(16)
	PrecMin	250	8	0	2	4	10	23	26	16	11	4	7	7	(17)
	PrecStd	74	7	13	10	14	22	24	31	29	29	33	7	10	(18)
	PrecStd	74	7	13	10	14	22	24	31	29	29	33	7	10	(18)
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.0	6.1	12.7	22.0	28.7	30.2	31.6	31.0	27.6	20.1	8.5	4.4	(19)
		MCWB	2.3	2.3	5.3	11.0	14.9	16.9	18.6	18.3	16.3	12.2	4.7	2.1	(20)
	2%	DB	2.2	3.1	9.3	17.9	24.8	27.7	29.2	28.3	23.8	15.3	5.4	1.8	(21)
		MCWB	0.5	0.6	3.9	8.4	13.0	15.8	17.9	17.5	15.0	8.9	2.4	0.1	(22)
	5%	DB	-0.7	0.6	6.9	15.3	22.4	25.4	27.3	26.0	21.0	12.7	3.3	-0.8	(23)
		MCWB	-2.0	-1.0	2.5	7.1	11.8	14.7	16.9	16.7	13.3	7.7	1.2	-1.9	(24)
	10%	DB	-4.5	-1.9	4.5	12.8	19.9	23.5	25.4	24.0	18.5	10.4	1.5	-3.4	(25)
		MCWB	-5.3	-3.2	1.3	5.8	10.6	14.1	16.4	15.8	12.4	6.3	0.1	-4.2	(26)
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.6	2.7	6.1	11.1	16.1	19.0	20.7	20.3	17.3	12.6	4.9	2.3	(27)
		MCDB	4.5	5.6	12.0	21.6	25.1	26.1	28.0	27.6	25.6	18.6	8.0	4.2	(28)
	2%	WB	0.5	0.8	4.1	9.0	14.1	17.2	19.1	18.7	15.7	9.8	2.8	0.2	(29)
		MCDB	2.1	3.0	8.6	16.5	22.8	23.6	25.7	25.2	22.4	13.9	5.0	1.6	(30)
	5%	WB	-2.0	-1.0	2.8	7.5	12.7	16.2	18.2	17.6	14.2	8.0	1.4	-1.9	(31)
		MCDB	-0.8	0.5	6.5	14.4	20.2	22.0	24.5	23.9	19.4	12.0	3.0	-0.8	(32)
	10%	WB	-5.3	-3.1	1.5	6.2	11.4	15.2	17.3	16.6	13.0	6.6	0.1	-4.2	(33)
		MCDB	-4.5	-2.0	4.3	12.2	18.6	21.4	23.3	22.4	17.2	10.1	1.6	-3.4	(34)
(36) Mean Daily Temperature Range	5% DB	MDBR	8.3	10.1	11.3	11.4	11.7	10.8	10.9	10.8	10.1	7.5	6.4	7.6	(35)
		MCDBR	11.3	12.5	12.5	15.0	15.2	14.1	14.0	14.3	14.9	11.8	8.1	9.5	(36)
		MCWBR	9.7	10.3	8.1	7.8	6.6	5.4	5.3	5.7	7.1	6.6	6.1	8.4	(37)
	5% WB	MCDBR	11.3	12.3	12.0	14.4	13.6	11.8	12.0	12.7	13.2	11.1	7.6	9.6	(38)
		MCWBR	9.8	10.2	8.0	7.9	6.3	5.2	5.3	5.7	6.9	6.6	5.9	8.6	(39)
(40) Clear-Sky Solar Irradiance	taub		0.246	0.270	0.297	0.338	0.358	0.382	0.397	0.367	0.332	0.301	0.261	0.234	(40)
	taud		2.229	2.278	2.291	2.364	2.383	2.328	2.345	2.400	2.497	2.534	2.362	2.161	(41)
	Ebn at Noon		652	794	863	879	879	857	835	838	824	764	640	548	(42)
	Edh at Noon		49	73	94	104	109	117	113	100	78	58	44	39	(43)
(44) All-Sky Solar Radiation	RadAvg		0.66	1.64	3.26	4.74	5.58	5.66	5.39	4.43	3.03	1.56	0.66	0.42	(44)
	RadStd		0.06	0.09	0.20	0.33	0.54	0.46	0.30	0.30	0.29	0.22	0.08	0.03	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	(e) +1.85	(f) N/A	(g) N/A	(h) -0.58	(i) -0.86	(j) N/A	(k) N/A	(l) N/A	(m) N/A
(47)	Regional (3 neighbors)	N/A	N/A	+1.63	N/A	-0.37	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page